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Vellore Institute of Technology
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School of Electronics Engineering (SENSE)

B. Tech – Electronics and Communication Engineering

**BECE320E – EMBEDDED C PROGRAMMING
PROJECT REPORT**

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REAL TIME ECG MONITORING, DISPLAY & DATA LOGGING SYSTEM

INTRODUCTION

Electrocardiography (ECG) is a technique used to monitor the electrical activity of the heart and identify abnormalities in cardiac function. Traditional ECG systems are often bulky, expensive, and limited to clinical environments, making continuous monitoring difficult.

With advancements in embedded systems, compact and portable ECG monitoring solutions have become feasible. This project presents a real-time ECG monitoring system using the ESP32-S3 microcontroller, AD8232 ECG sensor, a 0.96-inch OLED display, and Google Sheets integration. The system captures, processes, displays, and stores ECG data in real time, providing an efficient solution for heart monitoring.

NEED FOR THE PROJECT

Cardiovascular diseases are among the leading causes of death globally, increasing the need for continuous heart monitoring systems. Conventional ECG machines are not suitable for daily use due to their size and cost. There is a demand for portable, affordable, and easy-to-use monitoring systems that can be used outside hospitals.

This project addresses these challenges by providing a compact ECG monitoring system that enables real-time tracking and storage of heart activity. It promotes preventive healthcare by allowing early detection of abnormalities and maintaining historical data using cloud-based storage.

OBJECTIVES

The objective of this project is to design a real-time ECG monitoring system using the AD8232 sensor and ESP32 microcontroller. The system aims to acquire ECG signals, convert them into digital form using ADC, and process them to detect heartbeats.

Another objective is to calculate BPM using the R–R interval method and display both the ECG waveform and heart rate on an OLED display. Additionally, the

system is designed to upload ECG and BPM data to Google Sheets at regular intervals for storage and analysis.

COMPONENTS USED AND CONNECTIONS

The ESP32-S3 WROOM-1 microcontroller acts as the main processing unit. It reads analog ECG signals, processes them, and handles communication.

The AD8232 ECG sensor captures the electrical activity of the heart using three electrodes and outputs an analog signal.

The 0.96-inch OLED display is used for real-time visualization of ECG waveform and BPM using I2C communication.

The connections are as follows:

The OUTPUT pin of AD8232 is connected to GPIO 4 of ESP32, LO+ to GPIO 5, and LO- to GPIO 18. The OLED display is connected via I2C with SDA to GPIO 8 and SCL to GPIO 9. All components share a common ground and are powered using 3.3V.

PROCEDURE / WORKING

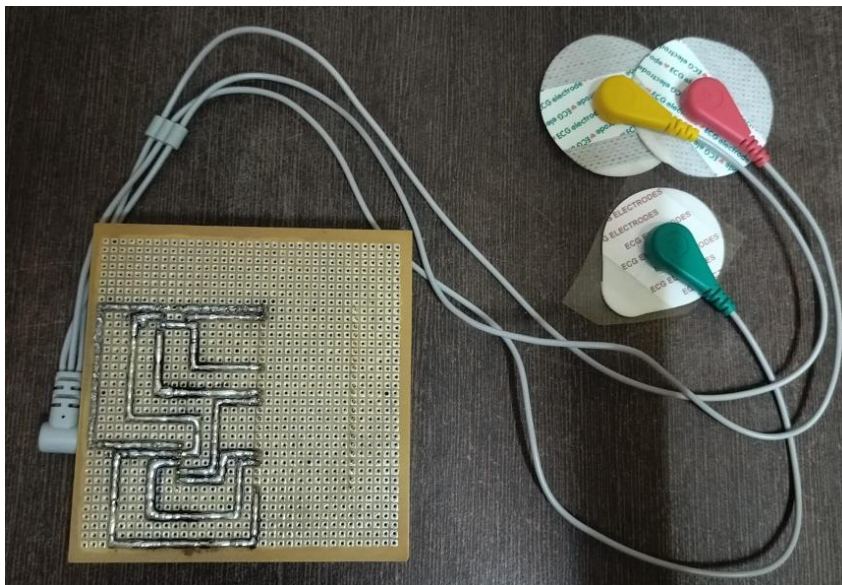
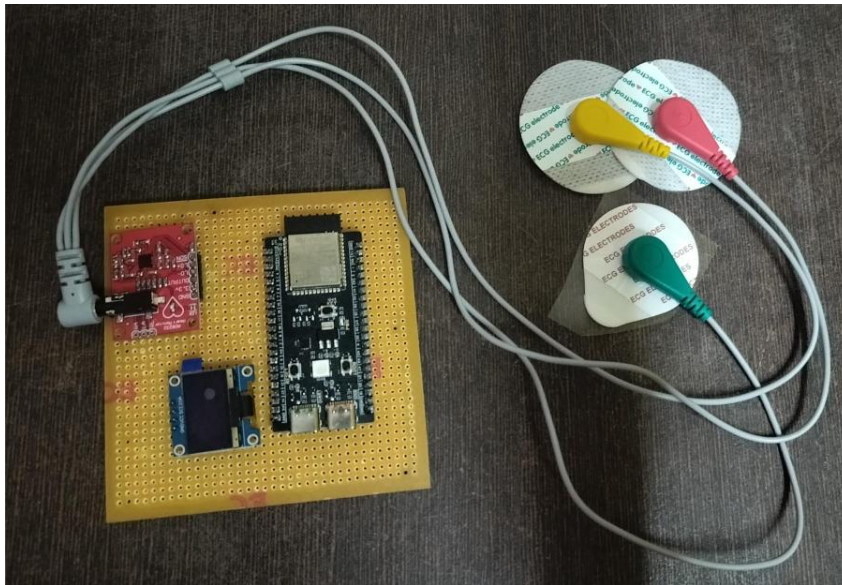
The working begins with placing electrodes on the body to capture ECG signals. The AD8232 sensor amplifies and filters the signal before sending it to the ESP32.

The microcontroller reads the signal using its 12-bit ADC and converts it into digital values.

The system detects R-peaks using a threshold-based method. The time interval between successive peaks is calculated, and BPM is determined using the formula $BPM = 60000 \text{ divided by the R-R interval}$.

The ECG waveform is displayed on the OLED, and the BPM value is updated in real time. Additionally, the ESP32 connects to WiFi and sends the ECG and BPM data to Google Sheets using HTTP requests.

This allows real-time data logging with timestamps, enabling remote monitoring and analysis of heart activity.



DEMO VIDEO LINK:

https://drive.google.com/file/d/1kJ051DtXFDjc3sY_5ywT_XoaR7H_lqyE/

CONCLUSION

The project successfully demonstrates a real-time ECG monitoring system using ESP32 and AD8232. The system accurately captures and processes ECG signals, calculates BPM using the R-R interval method, and displays the results in real time. The integration of Google Sheets enables data logging and remote monitoring, enhancing the functionality of the system. Overall, the project provides a cost-effective, portable, and efficient solution for basic heart monitoring and analysis.